

Rayane MOUHLI

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Ph.D. Student in Mathematics at Université Paris Cité (MAP5) & Sorbonne Université (LJLL)

EDUCATION

- **Ph.D. in Mathematics** 2023 - 2026
Université Paris Cité (MAP5) & Sorbonne Université (LJLL) Paris, France
 - PhD Thesis: "Ontogenesis via large deformations" under the supervision of Barbara Gris and Irène Kaltenmark.
- **Mathematics, Vision, Learning (MVA) - Research master's degree** 2022 - 2023
ENS Paris-Saclay Paris, France
 - Relevant Coursework: Geometric Data Analysis, Medical Image Analysis, Topological Data Analysis, Deep learning for medical image analysis, Geometry and shape spaces.
- **M.Sc., ENSAE engineering school** 2019 - 2023
Engineering Degree - Master Data Science, Statistics, and Machine Learning Palaiseau, France
 - Relevant Coursework: Advanced Machine Learning, Bayesian Statistics, Hidden Markov Models and Sequential Monte-Carlo, Natural Language Processing.
- **B.Sc., Lycée Claude Fauriel** 2016 - 2019
CPGE MPSI/MP Saint-Étienne, France
 - Intensive preparatory course for competitive entrance into top French engineering schools (Mathematics, Physics and Computer Science).

TEACHING EXPERIENCE

- **Teaching and Research Assistant (ATER)** 2026 - 2027
Université Paris Cité Paris, France
- **Teaching Assistant** 2023 - 2026
Université Paris Cité Paris, France
 - 1st-year Bachelor : Mathematics and Calculus 1, Mathematics and Calculus 2.
 - 3rd-year Bachelor: Differential Calculus and Dynamical Systems.
 - Supervision of projects and grading of defense presentations for 1st and 2nd year bachelor students.
- **Oral Examiner in Mathematics** 2022 - 2023
ICAM - Engineering School Paris, France
 - Examined weekly oral examinations in mathematics, in preparatory class for engineering school.
- **Data Analyst Instructor** 2021 - 2022
Noé Paris, France
 - Supervised practical sessions in Python and SQL applied to Product Management.

PUBLICATIONS

S=IN SUBMISSION

- [S.3] François A., Mouhli, R., Pierron, T. (2026). **A Framework for Joint Affine and Diffeomorphic Image Registration**. Preprint (submitted for publication).
- [S.2] Mouhli, R., Gris, B., Kaltenmark, I. (2026). **A Varifold-Based Score for Decoupling Deformations in Shape Analysis**. Preprint (submitted for publication).
- [S.1] Mouhli, R., Pierron, T. (2025). **Decoupling actions of finite-dimensional Lie groups and of groups of diffeomorphisms in the large deformation framework**. Preprint (submitted for publication).

SUPERVISION

- Antonin Lecocq, 2nd-year bachelor intern June 2026

ACADEMIC RESPONSIBILITIES

- **Ph.D. Student Representative of FP2M** 2024 - 2026
Fédération Parisienne de Modélisation Mathématique (FP2M) Paris, France
 - Organized inter-laboratory scientific events (MAP5, Modal'X, SAMM), promoted research work and managed the network of Ph.D. students within the federation.

TALKS & SEMINARS

• MAP5 Ph.D. Students Seminar <i>Introduction to shape analysis</i>	January 2026 Paris, France
• M2/Ph.D. Students Meeting at Sorbonne Université <i>Introduction to shape analysis</i>	January 2026 Paris, France
• Shape Seminar <i>Decoupling of vector fields using first variation of varifolds</i>	May 2025 Paris, France
• MAP5 Ph.D. Students Seminar <i>Decoupling of vector fields using first variation of varifolds</i>	February 2025 Paris, France
• Infinite-dimensional Geometry: Theory and applications <i>Decoupling of vector fields using first variation of varifolds</i>	February 2025 Vienna, Austria
• Congress of Young Researchers in Applied Mathematics (CJC-MA) <i>Decoupling of vector fields using first variation of varifolds</i>	October 2024 Lyon, France

PROFESSIONAL EXPERIENCE

• Research Internship <i>Scientific Computing and Imaging Institute - University of Utah</i> ◦ Medical image (MRI) registration via geodesic regression in the large deformation framework. ◦ Under the supervision of Dr. Sarang Joshi.	2023 (6 months) Salt Lake City, USA
• Research Internship <i>CEA</i> ◦ In-situ extraction of dislocation forests in molecular dynamics (skeleton extraction of point clouds).	2022 (6 months) Paris, France
• Data Scientist (R&D Internship) <i>fifty-five</i> ◦ Purchase probability prediction using deep learning and time series.	2021 - 2022 (6 months) Paris, France
• Data Analyst <i>FC Nantes eSport</i> ◦ Data processing, infographic creation, and data pipeline implementation.	2020 - 2021 Nantes, France
• Intern in the General Management Department <i>Métropole du Grand Paris</i> ◦ Contribution in the building of a management tool for a post-crisis recovery plan of the Métropole du Grand Paris.	2020 (2 months) Paris, France

SKILLS

- **Programming:** Python, SQL, $\text{\LaTeX}$
- **Languages:** English (C1+, 180+ Cambridge Linguaskill), Arabic (Beginner), Spanish (Beginner)